

Voice Forensics: Deepfake Detection Methodology

Hackathon Presentation Reference Guide

1. The Machine Learning Model

Which model did we use?

We utilized a fine-tuned version of **Wav2Vec2-XLSR** (Cross-Lingual Speech Representation), a state-of-the-art Transformer-based audio architecture developed by Meta, specifically trained for deepfake audio classification.

Why did we choose this particular model?

- **Cross-Lingual Robustness:** "XLSR" is trained on 53 languages. Because the hackathon targets Indian telephony (involving Hindi, English, and regional accents), we needed a model that would not misclassify an unfamiliar accent as "fake."
- **Raw Waveform Analysis:** Older models convert audio to images (spectrograms) before analyzing them, which blurs out microscopic details. Wav2Vec2 analyzes the raw audio wave directly, allowing it to detect fraction-of-a-millisecond glitches.

2. The Need for Machine Learning

Why use ML? Why not any other method?

To catch AI, you must use AI. Modern voice clones (like Edge-TTS or ElevenLabs) are so advanced that they perfectly mimic human breathing, pauses, and pitch variations. Classical algorithms (like simple volume thresholds) can no longer tell the difference. Machine learning is uniquely capable of finding the hidden, high-dimensional mathematical patterns created by the neural networks that generated the audio.

Does ML give the best accuracy?

Yes. In global forensic competitions (like the ASVspoof challenge), Transformer-based ML architectures like Wav2Vec2 absolutely dominate the leaderboards. They are currently the gold standard for audio deepfake detection.

3. The Mathematical Patterns We Detect

Our system uses a dual-approach: Classical Signal Processing (Macro-patterns) and Machine Learning (Micro-patterns).

A. Macro-Patterns (DSP Physical Measurements)

Before the ML model even runs, we measure physical acoustic properties of the wave:

- **Pause Rate & Breath Gaps:** AI generators often fail to breathe naturally. We measure silent gaps > 50ms per minute.
- **Spectral Cutoff (Hz):** Telephony audio drops dead at 4kHz or 8kHz. If the audio has pristine energy at 12kHz, it wasn't recorded on a real phone line—it was generated on a computer.
- **Noise Floor Flatness:** A real microphone records dynamic background noise. AI generates "perfect" digital silence between words, which is a mathematical dead giveaway.
- **F0 Curvature:** Real human pitch (loudness contour) varies constantly. AI voices often exhibit "unnatural stability," drifting less than a real human.

B. Micro-Patterns (The Machine Learning Artifacts)

The ML model looks for the invisible mathematical errors left behind by the AI generation process. There are about 6 main types of "Vocoders" (audio generation engines), and each leaves a unique math error:

- **HiFi-GAN & BigVGAN:** Leaves "*Checkerboard Periodicity*". Because the AI builds the audio by doubling the resolution over and over (transposed convolution), it leaves a microscopic repeating echo at exactly 8x, 4x, and 2x intervals.
- **Vocos / iSTFT:** Leaves a "*Phase Discontinuity*". It builds audio in chunks, causing a tiny mathematical mismatch at the exact boundary of every chunk.

- **EnCodec / DAC:** Leaves "*Quantization Noise*". Because it compresses audio into discrete digital tokens, it leaves a rigid 75Hz framing pattern that a human vocal cord could never physically produce.
- **Legacy AR (WaveNet):** Leaves an "*Over-smoothed Spectral Envelope*." The high-frequency sounds are too perfect and blurry, lacking the chaotic friction of real human breath.

4. Reverse-Engineering the Cloning Method (Layman's Terms)

Use this analogy when explaining to non-technical judges:

"Every voice clone is generated by a piece of software called a **Vocoder**. Even though there are dozens of voice cloning startups, almost all of them secretly reuse one of 6 main vocoder engine families under the hood.

Think of a vocoder like a computer printer. Even if a printed document looks flawless to the naked eye, different brands of printers (HP, Epson, Canon) leave microscopic, invisible tracking dots or ink splatters on the page.

Vocoders do the exact same thing to audio. Because of how they are coded, they leave invisible acoustic 'fingerprints.' Our AI model acts like a forensic microscope. It zooms into the sound wave, finds these invisible 'printer dots,' and reverse-engineers exactly which of the 6 engine families was used to generate the clone."